Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A sample of AlF_3 contains 3.0×10^{24} F^- ions. The number of formula unit of this sample are

(A) 9×10^{24}

(B) 3×10^{24}

(C) 0.75×10^{24}

(D) 1.0×10^{24}

Q.2

The density of $3M$ solution of sodium chloride is $1.252\ g\ mL^{-1}$. The molality of the solution will be: (molar mass, $NaCl = 585\ g\ mol^{-1}$)

(A) $2.60\ m$

(B) $2.18\ m$

(C) $2.79\ m$

(D) $3.00\ m$

Q.3

$10\ mL$ of $2M NaOH$ solution is added to $200\ mL$ of $0.5M$ of $NaOH$ solution. What is the final concentration?

(A) $0.57M$

(B) $5.7M$

(C) $11.4M$

(D) $1.14M$

Q.4

How many of $0.1\ N\ HCl$ are required to react completely with $1\ g$ mixture of Na_2CO_3 and $NaHCO_3$ containing equimolar amounts of two?

(A) $157.7\ mL$

(B) $15.77\ mL$

(C) $147.7\ mL$

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(D) $14.77\ \text{mL}$

Q.5

$100\ \text{cm}^3$ of 0.1NHCl is mixed with $100\ \text{cm}^3$ of 0.2 N NaOH solution. The resulting solution is

(A) $0.1\ \text{N}$ and the solution is basic

(B) $0.1\ \text{N}$ and the solution is acidic

(C) $0.05\ \text{N}$ and the solution is basic

(D) $0.05\ \text{N}$ and the solution is acidic

Q.6

What is the molarity of SO_4^{2-} ion in aqueous solution that contain 34.2ppm of $\text{Al}_2\left(\text{SO}_4\right)_3$?
(Assume complete dissociation and density of solution $1\ \text{g/mL}$)

(A) $3 \times 10^{-4}\text{M}$

(B) $2 \times 10^{-4}\text{M}$

(C) 10^{-4}M

(D) None of these

Q.7

$250\ \text{mL}$ of a sodium carbonate solution contains $2.65\ \text{g}$ of Na_2CO_3. If $10\ \text{mL}$ of this solution is diluted to one litre, what is the concentration of the resultant solution? (mol. wt. of $\text{Na}_2\text{CO}_3 = 106$)

(A) 0.1M

(B) 0.01M

(C) 0.001M

(D) 10^{-4}M

Q.8

A gaseous compound of nitrogen and hydrogen contains 12.5% (by mass) of hydrogen. The density of the compound relative to hydrogen is 16.
The molecular formula of the compound is:

(A) NH_2

(B) N_3H

(C) NH_3

(D) N_2H_4

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Q.9

The volume strength of $1.5\ NH_2O_2$ solution is

(A) 4.8

(B) 8.4

(C) 3.0

(D) 8.0

Q.10

In a mole of water vapour at STP, the volume actually occupied or taken by the molecules (i.e., Avogadro's No. $\times$ Volume of one molecule) is

(A) Zero

(B) Less than 1% of 22.4 litres

(C) About 10% of the volume of container

(D) 1% to 2% of 22.4 litres

Q.11

Dissolving $120\ g$ of a compound (mol. wt. 60) in $1000\ g$ of water gave a solution of density $1.12\ g/mL$. The molarity of the solution is:

(A) $1.00M$

(B) $2.00M$

(C) $2.50M$

(D) $4.00M$

Q.12

RH_2 (ion exchange resin) can replace Ca^{2+} ions in hard water as:

$$RH_2 + Ca^{2+} \longrightarrow RCa + 2H^+$$

If 1 L of hard water after passing through RH_2 has pH = 3, then hardness in parts per million of Ca^{2+} is :

(A) 10 ppm

(B) 40 ppm

(C) 100 ppm

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(D) 20 ppm

Q.13

If the density of CH_3OH is $0.80\ kg\ L^{-1}$, the volume of methanol to prepare $2.5\ L$ of $0.25M$ aqueous solution is

(A) $25.0\ mL$

(B) $32.0\ mL$

(C) $45.0\ mL$

(D) $56.0\ mL$

Q.14

The density of a 3 molal solution of sodium thiosulphate $(Na_2\ S_2O_3 \cdot 5H_2O)$ solution is $1.25\ g\ cm^{-3}$. The percentage by mass of sodium thiosulphate is

(A) 35.25%

(B) 39.20%

(C) 42.67%

(D) 46.67%

Q.15

The concentration of ethanol in the solution called 86-proof vodka is $6.5M$. If the density of the solution is $0.95\ g\ cm^{-3}$, the amount fraction of ethanol in vodka is

(A) 0.304

(B) 0.252

(C) 0.205

(D) 0.152

Answers & Solutions

Q.1 Answer:

1.0×10^{24}

Solution:

An, AlF_3 the number of F is 3 for one AlF_3 molecule $3\ F^- \equiv 1$ formula unit of AlF_3
$3.0 \times 10^{24}\ F^- \equiv \frac{1}{3} \times 3.0 \times 10^{24} AlF_3$ units

Q.2 Answer:

$2.79\ m$

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Solution:

The relation between molarity (M) **and molality** (m) **is**
$d=\mathrm{M}\left(\frac{1}{m}+\frac{\mathrm{M}_2}{1000}\right), \mathrm{M}_2=$ **Mol. mass of solute On putting value**
$1.252=3\left(\frac{1}{m}+\frac{58.5}{1000}\right)$
on solving $m=2.79$

Q.3 Answer:

$0.57\mathrm{M}$

Solution:

From molarity equation $\mathrm{M}_1\ \mathrm{V}_1+\mathrm{M}_2\ \mathrm{V}_2=\mathrm{MV}_{(\text{total})}$
$2\times\frac{10}{1000}+0.5\times\frac{200}{1000}=\mathrm{M}\times\frac{210}{1000}$
$120=\mathrm{M}\times 210$
$\mathrm{M}=\frac{120}{210}=0.57\mathrm{M}$

Q.4 Answer:

$157.7\ \mathrm{mL}$

Solution:

$\mathrm{Na_2CO_3}\ \mathrm{NaHCO_3}$
$\mathrm{x}\qquad(1-x)$
$\frac{x}{106}=\frac{1-x}{84}$ **given (moles are equal)**
$\mathrm{x}=0.557$
$\frac{0.557}{53}+\frac{0.443}{84}=\frac{\mathrm{V}\times 0.1}{1000}$
$\mathrm{V}=157.7\ \mathrm{mL}$

Q.5 Answer:

$0.05\ \mathrm{N}$ and the solution is basic

Solution:

Normality
$=\frac{\mathrm{N}_1\ \mathrm{V}_1-\mathrm{N}_2\ \mathrm{V}_2}{\mathrm{V}_1+\mathrm{V}_2}=\frac{0.2\times 100-0.1\times 100}{100+100}$
$=\frac{10}{200}=0.05\ \mathrm{N}\mathrm{NaOH}$

Q.6 Answer:

$3\times 10^{-4}\mathrm{M}$

Solution:

$\frac{\text{mass of } \mathrm{Al_2(SO_4)_3}}{\text{r water}}\times 10^6=34.2$
mass of water
litre solution contains $1000\ \mathrm{g}$ **of water** $\Rightarrow$ **In 1 litre solution, mass of**

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$\mathrm{Al_2\,(SO_4)_3} = \frac{34.2\times1000}{10^6} = 34.2\mathrm{mg}$

molarity of

$\mathrm{Al_2\,(SO_4)_3} = \frac{34.2\times10^{-3}}{342}\mathrm{M} = 10^{-4}\mathrm{M}$

$\mathrm{Al_2\,(SO_4)_3\,(aq)} \longrightarrow 2\mathrm{Al^{3+}(aq)} + 3\mathrm{SO_4^{2-}(aq)}$

$10^{-4}\mathrm{M} \qquad\qquad 2\times10^{-4}\mathrm{M} \quad 3\times10^{-4}\mathrm{M}$

$\left[\mathrm{SO_4^{2-}}\right] = 3\times10^{-4}\mathrm{M}$

Q.7 Answer:

$0.001\mathrm{M}$

Solution:

Molarity

$= \frac{\mathrm{Wt}\times1000}{\mathrm{MW}\times\mathrm{V}} = \frac{2.65\times1000}{106\times250} = 0.1\mathrm{M}$

$\mathrm{M_1\,V_1} = \mathrm{M_2\,V_2}$

$\therefore\ 10\times0.1 = 1000\times\mathrm{M_2} = 0.001\mathrm{M}$

Q.8 Answer:

$\mathrm{N_2H_4}$

Solution:

In an unknown compounds containing N **and** H **given** $\%$ **of** $\mathrm{H} = 12.5\%$

$\therefore \quad \%$ **of** $\mathrm{N} = 100 - 12.5 = 87.5\%$

\begin{array}{|c|c|c|c|}

**\hline Element & Percentage & Atomic ratio & Simple ratio **

**\hline \mathrm{H} & 12.5 \% & \frac{12.5}{1}=12.5 & \frac{12.5}{6.25}=2 **

**\hline \mathrm{N} & 87.5 & \frac{87.5}{14}=6.25 & \frac{6.25}{6.25}=1 **

\hline

\end{array}

$2\times$ **vapour density** $=$ **Mol.** $\mathrm{wt} = \mathrm{molwt.}$

$= 16\times2 = 32$

Molecular formula $= n\times$ **empirical formula**

mass $n = \frac{32}{16} = 2$

$\therefore$ **Molecular formula of the compound will be** $= (\mathrm{NH_2})_2 = \mathrm{N_2H_4}$

Q.9 Answer:

8.4

Solution:

Normality of 10 volume $\mathrm{H_2O_2} = 1.78\ \mathrm{N}$

Thus, volume strength of $1.78\ \mathrm{N}$ **solution** $= 10$

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∴ Volume strength of

$1.5\ NH_2O_2 = 1.5 \times \frac{10}{1.78} = \frac{15}{1.78} = 8.4$

Q.10 Answer:

Less than 1% of 22.4 litres

Solution:

IT is about $0.224L$

Q.11 Answer:

$2.00\mathrm{M}$

Solution:

Given mass of solute $(w) = 120$ g **mass of solvent** $(w) = 1000$ g **Mol. mass of solute** $= 60$ g **density of solution** $= 1.12$ g/ml **From the given data, Mass of solution** $= 1000 + 120 = 1120$ g $\because \quad d = \frac{\text{Mol. mass}}{V}$ **or** $V = \frac{\text{Mol. mass}}{d}$ **Volume of solution** $V = \frac{1120}{1.12} = 1000\text{ml}$ **or** $= 1$ **litre Now molarity** $(\mathrm{M}) = \frac{\mathrm{W}}{\text{Mol. mass} \times V(\text{ lit })} = \frac{120}{60 \times 1} = 2\mathrm{M}$

Q.12 Answer:

20 ppm

Solution:

[H^+] =10^{-3} M [∵ pH = 3 = - log [H^+]

=10^{-3} moles per litre

[Ca^{2+}] = 5 × 10^{-4} moles per litre

= 5 × 10^{-1} moles of Ca^{2+} per 10^3 L

= 5 × 10^{-1} × 40 g Ca^{2+} per 10^6 g of H_2O

Ca^{2+} = 20 ppm = (H).

Q.13 Answer:

$25.0\ \mathrm{mL}$

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Solution:

Since molarity, $M = n_2/V$**, we have Amount of** CH_3OH **required,** $n_2 = MV = \left(0.25 \text{ mol L}^{-1}\right)(2.5 \text{ L}) = 0.625 \text{ mol}$

Mass of CH_3OH **required,** $m_2 = n_2 M_2 = (0.625 \text{ mol})\left(32 \text{ g mol}^{-1}\right) = 20.0 \text{ g} = 20.0 \times 10^{-3} \text{ kg}$

Volume of CH_3OH **required,** $V = \frac{m_2}{\rho} = \frac{20.0 \times 10^{-3} \text{ kg}}{0.80 \text{ kg L}^{-1}} = 0.025 \text{ L} = 25.0 \text{ mL}$

Q.14 Answer:

42.67%

Solution:

Molar mass of $Na_2\ S_2O_3 \cdot 5H_2O$ **is** $M = (2 \times 23 + 2 \times 32 + 3 \times 16 + 5 \times 18)\text{gmol}^{-1} = 248 \text{ g mol}^{-1}$

The mass of 3 mol of $Na_2\ S_2O_3 \cdot 5H_2O$ **is** $m_2 = (3 \text{ mol})\left(248 \text{ g mol}^{-1}\right) = 744 \text{ g}$

The mass of solution will be $m = m_1 + m_2 = 1000 \text{ g} + 744 \text{ g} = 1744 \text{ g}$

Mass percentage of sodium thiosulphate $= \frac{m_2}{m} \times 100 = \left(\frac{744 \text{ g}}{1744 \text{ g}}\right) \times 100 = 42.67\%$

Q.15 Answer:

0.152

Solution:

For 1 L **of vodka, we have** $n_2 = 6.5 \text{ mol}$ **and** $V = 1000 \text{ cm}^3$ **Mass of solution,** $m = V\rho = \left(1000 \text{ cm}^3\right)\left(0.95 \text{ g cm}^{-3}\right) = 950 \text{ g}$

Mass of ethanol, $m_2 = n_2 M_2 = (6.5 \text{ mol})\left(46 \text{ g mol}^{-1}\right) = 299 \text{ g}$

Mass of water, $m_1 = m - m_2 = 950 \text{ g} - 299 \text{ g} = 651 \text{ g}$

Amount of water, $n_1 = \frac{m_1}{M_1} = \frac{651 \text{ g}}{18 \text{ g mol}^{-1}} = 36.167 \text{ mol}$

Amount fraction of ethanol, $x_2 = \frac{n_2}{n_1 + n_2} = \frac{6.5}{(36.167 + 6.5)} = 0.152$

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